**Printed Pages: 02** **Sub Code: BOP-232**

**Paper Id: 233442** **Roll No.** | | | | | | | | | | |

**B. PHARM.**
**(SEM III) THEORY EXAMINATION 2022-23**
**PHARMACEUTICS-II (UNIT OPERATIONS)**

***Time: 3 Hours*** ***Total Marks: 70***

**Note:** Attempt all Sections. If require any missing data; then choose suitably.

**SECTION A**

**1. Attempt *all* questions in brief.** **2 x 7 = 14**

(a) Describe Unit Process with suitable examples.
(b) What are Derived Units? Give example.
(c) Differentiate between Surface Filtration & Depth Filtration.
(d) Compare Raw Water & Hard Water.
(e) Define CMC and FMC.
(f) How heat is transferred through conduction & convection?
(g) Give two mechanisms of drying.

**SECTION B**

**2. Attempt any *three* of the following:** **7x3=21**

(a) Compare & contrast Open Loop and Closed Loop Control system in details with examples.

(b) Discuss about factors affecting Filtration? Draw a neat sketch of Metafilter and describe its principle, construction and working.

(c) Categorize the types of Dryers? Describe the construction, working and principle of Tray Dryer.

(d) Explain the term Refrigerant with examples? Describe Refrigeration Cycle with a neat and labeled diagram.

(e) Classify material of construction. Write a descriptive note on glasses used in pharmaceutical industry. Add a note on general composition of glass.

**SECTION C**

**3. Attempt any *one* part of the following:** **7x1=10**

(a) Explain the terms Steady State & Unsteady State. Add a note on Equilibrium State and Rate Process.

(b) Write a descriptive note on
(i) Temperature Measurement
(ii) Level Measurement.

**4. Attempt any *one* part of the following:** **7x1=10**

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(a) What do you mean by Water for Injection? Illustrate the method of preparation of WFI on industrial scale?

(b) Describe the principle, construction, working, uses, advantages & disadvantages of Perforated basket Centrifuge.

**5. Attempt any *one* part of the following: 7x1=10**

(a) Discuss about the pharmaceutical applications of Drying? Explain the principle, construction and working of Fluidized Bed Dryer.

(b) Draw a well labeled diagram of Freeze Dryer. Explain its principle, construction, working & applications also.

**6. Attempt any *one* part of the following: 7x1=10**

(a) What is Air Conditioning? Describe the approaches for achieving air conditioning.

(b) Summarize the components of Psychometric Chart. Describe its significance and important features in pharmacy.

**7. Attempt any *one* part of the following: 7x1=10**

(a) Define Corrosion? Classify the types of corrosion. How corrosion can be prevented?

(b) Write a descriptive note on i) Mechanical Records ii) Fire Hazards

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